



**Beijing-Dublin International College**



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**SEMESTER II FINAL EXAMINATION - 2018/2019**

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**Beijing-Dublin International College**

**《UNIVERSITY PHYSICS 1》 BDIC1015J**

Hao ZHU and Boyang LIU

**Time Allowed: 95 minutes**

**Instructions for Candidates**

Full marks will be awarded for complete answer to All questions.

**BJUT Student ID:** \_\_\_\_\_ **UCD Student ID:** \_\_\_\_\_

I have read and clearly understand the Examination Rules of both Beijing University of Technology and University College Dublin. I am aware of the Punishment for Violating the Rules of Beijing University of Technology and/or University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I accept the punishment thereof.

**Honesty Pledge:** \_\_\_\_\_ **(Signature)**

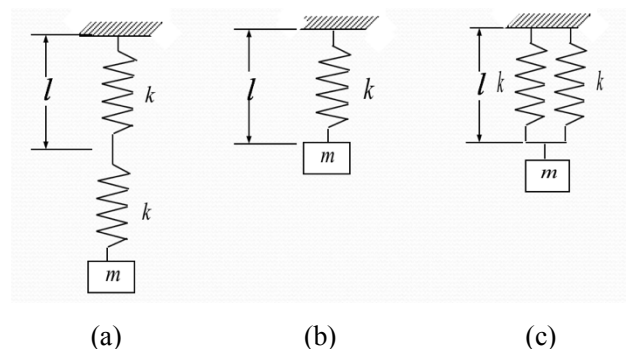
**Instructions for Invigilators**

Non-programmable calculators are permitted.  
No rough-work paper is to be provided for candidates.

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**Part 1: Multiple Choice (3 marks each question, 30 marks altogether)**

- An electron moving along the  $x$ -axis has a position given by  $x = 16te^{-t}$ (m), where  $t$  is in seconds. How far is the electron from origin when it temporarily stops?
  - $16e$
  - $16/e$
  - $e^{16}$
  - $e^{-16}$
- A 2kg object is subjected to three forces that give it an acceleration  $\vec{a} = -(8 \text{ m/s}^2)\vec{i} + (6 \text{ m/s}^2)\vec{j}$ . If two of the three forces are  $\vec{F}_1 = (30\text{N})\vec{i} + (16\text{N})\vec{j}$  and  $\vec{F}_2 = (-12\text{N})\vec{i} + (8\text{N})\vec{j}$ . Try to find the third force?
  - $(-34\text{N})\vec{i} - (12\text{N})\vec{j}$
  - $(34\text{N})\vec{i} + (12\text{N})\vec{j}$
  - $(34\text{N})\vec{i} - (12\text{N})\vec{j}$
  - None of above
- A 91kg man lying on a surface of negligible friction throws a 68g stone away from himself, giving it a speed of 4.0 m/s. What speed does the man acquire as a result?
  - $-1.5 \times 10^{-3} \text{ m/s}$
  - $-3.0 \times 10^{-3} \text{ m/s}$
  - $-4.5 \times 10^{-3} \text{ m/s}$
  - $-6.0 \times 10^{-3} \text{ m/s}$
- A father racing his son has half the kinetic energy of the son, who has half the mass of the father. The father speeds up by 1.0 m/s and then has the same kinetic energy as the son. What are the original speeds of the father ( $v_f$ ) and the son ( $v_s$ ) respectively?
  - $v_f = 1.2 \text{ m/s}, v_s = 2.4 \text{ m/s}$
  - $v_f = 2.4 \text{ m/s}, v_s = 1.2 \text{ m/s}$
  - $v_f = 2.4 \text{ m/s}, v_s = 4.8 \text{ m/s}$
  - $v_f = 4.8 \text{ m/s}, v_s = 2.4 \text{ m/s}$
- Several forces acting on a rigid body with a fixed axis simultaneously. If  $\sum_i \vec{F}_i = 0$ , then the rigid body [].
  - will not rotate.
  - the angular speed will not change
  - the angular speed will change
  - None of above
- As shown in figure, there are three oscillating systems (a), (b) and (c). Their angular frequencies are  $\omega_a$ ,  $\omega_b$  and  $\omega_c$ , respectively. What is the ratio of  $\omega_a : \omega_b : \omega_c$ ?
  - 1: 1: 2
  - $2: \sqrt{2}: 1$
  - $1: \sqrt{2}: 2$
  - $2: 1: 1/2$



7. The function of a simple harmonic motion is  $y = A\cos(Bt - Cx)$ , where  $A$ ,  $B$  and  $C$  are positive constants. Which one of the following descriptions is correct?

- (A) The wave speed is  $C$ .
- (B) The period is  $1/B$ .
- (C) The wavelength is  $2\pi/C$ .
- (D) The angular frequency is  $2\pi/B$ .

8. In the relationship  $p + {}^{16}_9\text{F} \rightarrow \alpha + {}^{16}_8\text{O}$ , the masses are  $m(p) = 1.007825\text{u}$ ;  $m(\alpha) = 4.002603\text{u}$ ;  $m(\text{F}) = 18.998405\text{u}$ ;  $m(\text{O}) = 15.994915\text{u}$ , where  $u$  is called atomic mass unit. Try to calculate the energy released from this reaction. ( $u = 1.66 \times 10^{-27}\text{kg}$ ,  $c = 3 \times 10^8 \text{ m/s}$ )

- (A)  $1.3 \times 10^{-12} \text{ J}$
- (B)  $1.4 \times 10^{-12} \text{ J}$
- (C)  $1.5 \times 10^{-12} \text{ J}$
- (D)  $1.6 \times 10^{-12} \text{ J}$

9.  $v_p$ ,  $\bar{v}$  and  $\sqrt{\overline{v^2}}$  denote the most probable speed, the average speed and the root-mean square speed respectively. For the same ideal gas at equilibrium state, which of the following statements is correct?

- (A)  $v_p < \bar{v} < \sqrt{\overline{v^2}}$
- (B)  $\bar{v} < v_p < \sqrt{\overline{v^2}}$
- (C)  $\sqrt{\overline{v^2}} < v_p < \bar{v}$
- (D)  $\bar{v} < \sqrt{\overline{v^2}} < v_p$

10. The first law of thermodynamics says the change in internal energy of a system is equal to the sum of the work done on it and the amount of heat energy transferred to it ( $dE = dQ - dW$ ). Which of the following statements is INCORRECT?

- (A) For isochoric process,  $dW = 0$ ,  $dQ = dE$
- (B) For isobaric process,  $dE = dQ - dW$
- (C) For the isothermal process,  $dQ = 0$ ,  $dE = -dW$
- (D) For adiabatic process,  $dQ = 0$ ,  $dE = -dW$

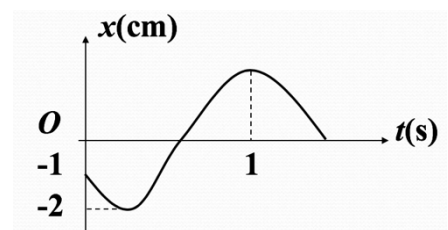
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**Part2:** Blank Filling (20 marks altogether)

11. (2 marks) A wheel with a fixed axle is rotating and the angular velocity is given as a function of time by  $\omega = 2t + 4t^2$ . Then the angular acceleration at  $t = 0.5 \text{ s}$  is \_\_\_\_\_.

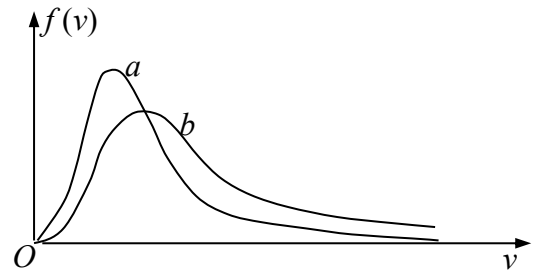
12. (6 marks) An Earth satellite with mass  $m$  moves on a circular orbit with radius  $3R_E$ , where  $R_E$  is the radius of the Earth, and mass of the Earth is known to be  $M_E$ . In this case, the kinetic energy of the satellite is \_\_\_\_\_; The gravitational potential energy of the satellite is \_\_\_\_\_.

13. (4 marks) An oscillation curve of a simple harmonic oscillator is as shown in figure. The equation of simple harmonic motion is \_\_\_\_\_ (cm);



14. (4 marks) If the kinetic energy of a particle is  $4m_0c^2$ , then the corresponding relativistic mass is \_\_\_\_\_. If the total energy of a particle is  $4m_0c^2$ , the corresponding relativistic mass is \_\_\_\_\_. ( $m_0$  is the rest mass of the particle)

15. (4 marks) The two curves  $a$  and  $b$  show the speed distributions of hydrogen gas and oxygen gas at temperature  $T$ . Curve  $b$  is the speed distribution of the \_\_\_\_\_ gas. The ratio of the most probable speeds of the two ideal gases is  $v_p(H_2)/v_p(O_2) = \underline{\hspace{2cm}}$ .



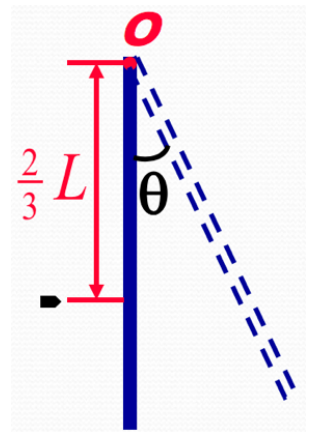
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**Part 3: Calculation**(10 marks each question, 50 marks altogether)

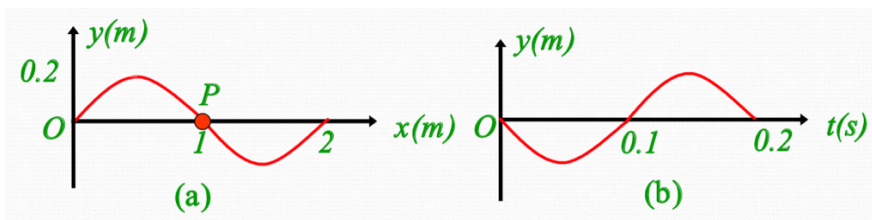
16. The velocity  $\vec{v}$  of a particle moving in the 'xy'-plane is given by  $\vec{v} = (6t - 4t^2)\vec{i} + 8\vec{j}$ , with  $\vec{v}$  in meters per second and  $t (>0)$  in seconds.

- What is the acceleration when  $t = 3$  s?
- When (if ever) is the acceleration zero?
- When (if ever) is the velocity zero?
- When (if ever) does the speed equal 10 m/s?

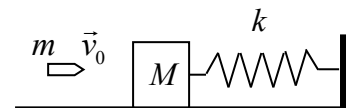
17. A thin and uniform rod of length  $L$  and mass  $M$  can rotate without friction about horizontal axis  $O$  which passes through the top of the rod. A bullet with mass  $m$  is shot to the point with distance  $2L/3$  away from the pivot. The bullet is shot in the rod with velocity  $\vec{v}_0$ . The rotational inertia of the rod about  $O$  is  $J = mL^2/3$  and  $M = 4m$ . Find the maximum deflection angle of the rod and the bullet with respect to the vertical axis.



18. The wave form of  $t = 0$  and the oscillation curve of point  $P$  are shown in the diagrams below. What is the function of the wave?



19. A block of mass  $M$ , at rest on a horizontal, frictionless table, is attached to a rigid support by a spring of force constant  $k$ . A bullet of mass  $m$  and speed  $v_0$  strikes the block as shown in figure. The bullet remains embedded in the block. Determine the equation of the resulting simple harmonic motion, in terms of  $m$ ,  $M$ ,  $v_0$  and  $k$ .



20. The total energy of a meson moving at a very high speed is  $E = 2000\text{MeV}$ , and the rest energy of this meson is  $E_0 = 100\text{MeV}$ . How long can the meson move if the proper time of the meson is  $\tau_0 = 2 \times 10^{-6}$  s?